Batu Şengezer

Embedded Software Engineer & Edge AI Researcher

Real-Time Safety-Critical Embedded Systems | Edge AI | Applied ML/DL | Hardware-Software Co-Design | Power-Aware Computing | Computer Vision | FPGA/Hardware Acceleration | Multi-Agent Coordination

Ankara, Turkey

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SUMMARY

Electrical and Electronics Engineer working at the intersection of real-time embedded systems and edge AI. Currently an Embedded Software Engineer at ASELSAN, developing and verifying mission-critical, real-time C/C++ software for guidance and navigation systems. Independently designs and executes hardware-grounded research on deploying and characterizing machine learning inference across constrained platforms (microcontrollers, embedded GPUs, FPGAs), measuring the real costs of acceleration in latency, memory, and power. Seeking graduate research (MASc) in embedded systems, edge AI, and hardware-software co-design.

EDUCATION

2024–2026 **Bilkent University**, Faculty of Business Administration — Ankara, Turkey
Master of Business Administration (MBA) — CGPA: 4.0/4.0 (100%), Ranked 1st in graduating cohort

2017–2021 **Bilkent University**, Faculty of Engineering — Ankara, Turkey
B.Sc. in Electrical and Electronics Engineering — CGPA: 2.98/4.0 (85%)
Relevant coursework: Microprocessors, Signals & Systems, Control Systems, Digital Circuit Design, Statistical Learning & Data Analytics, Machine Learning, Algorithms & Programming, Computer Networks

2013–2017 **TED Ankara College Foundation Schools** — Ankara, Turkey
International Baccalaureate (IB) Diploma — Grade: 94.24/100

RESEARCH & TECHNICAL PROJECTS

Edge AI Across the Compute Spectrum — *Independent Research, 2026,* github.com/batu-sengezer

Designed and executed a three-tier study deploying an identical 38K-parameter neural network classifier across a microcontroller, an embedded GPU, and a general-purpose CPU/GPU/NPU platform to measure the real cost and benefit of hardware acceleration for small ML workloads.

- **Tier 1 (STM32F407, ARM Cortex-M4, FreeRTOS):** trained the classifier myself, then deployed it as an INT8-quantized model (11.3× smaller than FP32, 0.17% accuracy loss) across three concurrent RTOS tasks; used a hardware cycle counter (DWT) to measure a maximum observed inference latency of 786.89 µs over 1,000 runs under scheduler contention, at 0.79% CPU utilization.
- **Tier 2 (NVIDIA Jetson Orin Nano, CUDA/TensorRT):** benchmarked TensorRT inference, a handwritten tiled CUDA matrix-multiplication kernel (up to 266× over a single-threaded CPU baseline), a live camera vision pipeline, latency under synthetic CPU/memory contention, and power-aware efficiency analysis across power modes (a 15W cap improved inferences-per-watt by 21% over the uncapped mode).
- **Tier 3 (Apple M5 Pro, PyTorch/CUDA/Core ML):** compared CPU, GPU (Metal), and Neural Engine execution; found GPU throughput was 6.5× *slower* than CPU at batch size 1 but 2.31× *faster* at batch 1024, and used independent power-rail measurements to confirm the Neural Engine was never engaged by the OS scheduler for this workload.
- Synthesized all three tiers into a cross-platform latency/throughput/energy-efficiency comparison; concluded that acceleration benefit is a property of arithmetic intensity per call, and that the smallest platform tested already met real-time and power budgets with wide margin.

Decentralized Learning for Coordination in Energy-Constrained Autonomous Agent Swarms — *Research with Prof. Fehmi Tanrısever, Bilkent Faculty of Business Administration, in progress*

Researching multi-agent federated learning for coordinating swarms of autonomous agents under energy constraints, where agents learn to communicate only when the expected benefit of communication exceeds its energy cost. Preprint in preparation.

Undergraduate Capstone: Portable Diagnostic System for Arçelik Refrigerators (ServiceKIT) — *Bilkent–Arçelik Industry Project, 2020–2021*

- Built an IoT field diagnostic and firmware-update kit centered on a Raspberry Pi, presented at the Bilkent EEE Industrial Design Projects Fair 2021; interfaced sensors (temperature, vibration, control-card telemetry) and built a custom failure-mode dataset to train a Scikit-Learn anomaly/fault-detection classifier (e.g., compressor, fan).
- Deployed the trained model to a cloud backend for field diagnostics and implemented over-the-air firmware updates to the refrigerator's electronic control card.

FPGA-Based Cargo Distribution Conveyor System — *Digital Design Course Project, 2018*

Designed and implemented a fully FPGA-native control system for a two-belt cargo conveyor on a Xilinx Artix-7 (Basys 3), with ultrasonic sensor pulse/echo timing and DC motor control expressed entirely as synthesizable hardware logic.

PROFESSIONAL EXPERIENCE

01/2026–Present	**Embedded Software Engineer, ASELSAN Inc.** Development, verification, and implementation of mission-critical, real-time embedded software (C/C++) for guidance and navigation systems.	Ankara, Turkey
03/2024–12/2025	**Project Manager, ASELSAN Inc.** Technical delivery lead for airborne navigation systems and inertial measurement units (IMUs), coordinating hardware, software, and systems engineering teams within the Avionics, Guidance and Navigation (AGS) Business Sector.	Ankara, Turkey
01/2023–03/2024	**Project Engineer, ASELSAN Inc.** Technical and schedule coordination across multi-disciplinary teams for airborne navigation systems and IMU development on national defense platforms.	Ankara, Turkey
12/2021–12/2022	**Avionics Hardware Design Engineer, ASELSAN Inc.** Digital hardware design and testing for avionic systems (cockpit display units, flight control computers) across helicopter and fighter jet platforms.	Ankara, Turkey
03/2021–11/2021	**Software Engineer, Arçelik A.Ş.** Development of embedded software and computer vision/image processing algorithms for the R&D Department (Dishwasher Plant).	Ankara, Turkey
07/2020–08/2020	**Engineering Intern, Arçelik A.Ş.** Developed computer vision software (Python, OpenCV) for quality control on refrigerator production lines.	Ankara, Turkey

PROFESSIONAL PROJECTS

Guidance Kit Development Projects — *ASELSAN Inc.*

Developed embedded software (C/C++) across multiple guidance kit subsystems, such as laser seeker, motor control, and anti-jam antenna integration, plus a desktop interface application and a Java-based Ground Control Station.

Airborne Navigation Systems Development & Delivery — *ASELSAN Inc.*

Simultaneously managed 3 complex projects (combined value over $9M USD) delivering airborne navigation systems and IMUs for national platforms, including the KAAN fighter jet, coordinating multidisciplinary teams of 40+ engineers.

Airborne Electronics Hardware Development — *ASELSAN Inc.*

Contributed to hardware development lifecycles for avionic units (cockpit display units, flight control computers), covering design, verification, and maintenance work packages for military aircraft.

TECHNICAL SKILLS

Languages	C, C++, Python, CUDA, VHDL, Java, SQL
ML & Edge AI	TensorFlow/Keras, PyTorch, TensorRT, Core ML, Scikit-Learn, OpenCV, NumPy/Pandas, Matplotlib, Quantization (INT8/FP16), Computer Vision and Machine Learning
Embedded & RTOS	Embedded Software Development, Object-Oriented Programming (OOP), Software Design Patterns, Real-Time Operating Systems (RTOS), FreeRTOS, Micrium OS (μC/OS), Memory Management, ARM Cortex-M/A, STM32, Jetson (JetPack/L4T), DWT Cycle-Accurate Timing, X-CUBE-AI, Oscilloscope/Multimeter Measurements
Hardware & EDA	Hardware Development, Digital Hardware Design, FPGA (Xilinx/Basys 3), Altium Designer, Mentor Graphics, LTSpice, MATLAB/Simulink
Tools	Git, Tortoise SVN, Linux/SSH, IBM Rational DOORS, Jira, Primavera P6, MS Project, SAP ERP

CERTIFICATIONS

- Professional Scrum Master I (PSM I) — Scrum.org
- Certified Associate in Project Management (CAPM) — PMI
- Bid and Proposal Management Foundation (CF APMP) — APMG

LANGUAGES

Turkish (Native), English (IELTS Academic 7.5 Overall), German (Elementary Proficiency)